Technical Meeting on the INNOVECMO Project

Context and Preliminary Summary

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Abstract

This document reports and synthesizes the outcomes of a series of technical and clinical meetings held in Valladolid in February 2026, within the context of the development of an alternative, valveless pumping concept for extracorporeal membrane oxygenation (ECMO). The activities comprised a laboratory visit at the University of Valladolid, a post-laboratory strategic meeting focused on future research directions, and a clinically oriented discussion at Hospital Clínico Universitario de Valladolid.

The laboratory visit provided direct insight into experimental implementations of Liebau-based, wave-driven pumping mechanisms, including the fabrication of highly compliant tubes, displacement-controlled actuation strategies, and associated pressure, flow, and imaging instrumentation. Building on these observations, a staged experimental–numerical roadmap was defined, emphasizing progressive complexity, robust treatment of large deformations and contact, and the coupling of high-fidelity fluid–structure interaction models with reduced-order representations.

The clinical meeting anchored the research objectives to current ECMO practice by identifying realistic operating envelopes, including typical adult flow rates, pressure rises, venous suction limits, cannula dimensions, circuit volumes, and transient operating procedures. Current centrifugal ECMO pumps were explicitly identified as the technological baseline, together with their known limitations related to hemolysis, thrombogenesis, and sensitivity to low-flow and high-shear regimes.

By integrating experimental feasibility, numerical strategy, and clinical constraints, the document establishes a coherent framework for assessing the potential of valveless, wave-driven pumping as a viable alternative to rotary ECMO pumps. The conclusions define clear design targets and modeling requirements, providing a solid foundation for subsequent experimental validation, reduced-order model development, and system-level analysis.

1 Introduction and Context

This document provides the contextual framework for a technical coordination meeting held in Valladolid on February 2nd, 2026, within the scope of the **INNOVECMO** research project. The purpose of the meeting was to align experimental, numerical, and control-oriented activities related to the development and validation of an innovative *valveless Liebau-based pump* for Extracorporeal Membrane Oxygenation (ECMO) systems.

The meeting brought together researchers from **CIMNE** (International Center for Numerical Methods in Engineering) and the **University of Valladolid (UVA)**, with the objective of

consolidating a shared technical vision across work packages, in particular those related to experimental validation, high-fidelity numerical modeling, and reduced-order modeling for design and control.

From the CIMNE side, the meeting was attended by Joaquín A. Hernández and Eduardo Soudah, who are responsible for the development of the computational framework underpinning the project. This framework includes fluid–structure interaction (FSI) modeling, contact mechanics for tube deformation, and the construction of projection-based reduced-order models aimed at enabling efficient design exploration and real-time control strategies.

The University of Valladolid delegation included José Sierra, Francisco Castro, and other collaborators involved in experimental setup design, ECMO emulation, and clinical relevance assessment. A central topic of discussion was the integration of the Liebau pump into an ECMO emulator capable of reproducing physiologically relevant flow and pressure conditions, as well as the identification of measurable quantities suitable for model validation and control.

In addition to the technical meeting, the participants also visited the **Hospital Clínico Universitario de Valladolid** in order to gain direct insight into clinical ECMO practice and constraints. The outcomes of this visit, and their implications for the project, will be documented and incorporated in a subsequent update of this report.

The present document serves as an introductory reference and will be progressively expanded to include a structured narrative derived from the audio recordings of the meeting, translated and consolidated into a coherent technical account.

2 Discussion Summary I: Conceptual Background and Technical Challenges

2.1 Research Background and Methodological Framework

The discussion opened with a technical overview of the research trajectory of the Valladolid group, whose work since the early 2000s has progressively shifted from classical fluid mechanics toward strongly coupled fluid–structure interaction (FSI) problems involving large deformations, nonlinear constitutive behavior, and dynamic energy exchange between fluid and solid domains. Early modeling approaches, typically based on rigid-wall or weakly coupled assumptions, were identified as insufficient for capturing the dominant mechanisms governing wave-driven transport in compliant conduits.

A defining aspect of the group's methodology is the construction of a unified computational–experimental framework, in which experimental observations are not merely used for validation, but to extract dominant deformation, contact, and wave-propagation patterns that can be systematically embedded into numerical models. This philosophy underpins the subsequent discussion on reduced-order modeling and control-oriented simulation.

2.2 Liebau-Based Pumping Principle

The pumping concept was traced to the Liebau principle, originally formulated in the mid-20th century, whereby periodic external compression of a flexible tube generates a net directional

flow in the absence of valves. From a physical standpoint, the mechanism is governed by the excitation, propagation, and reflection of pressure and deformation waves within a compliant fluid-filled conduit.

Early considerations of implantable configurations were quickly deemed impractical, leading to a focus on extracorporeal implementations. In this context, the Liebau-based pump presents several intrinsic advantages over rotary devices, most notably the elimination of high shear regions associated with rotating blades, which are known to exacerbate hemolysis and platelet activation in blood-contacting applications such as ECMO.

2.3 Wave Dynamics, Asymmetry, and Resonant Operation

A central technical point emphasized throughout the discussion was that the pump should be interpreted as a wave-dominated system rather than as a localized displacement device. Periodic excitation induces pressure and wall-deformation waves that travel along the tube and undergo partial reflection at geometric discontinuities and boundary interfaces.

Due to spatial asymmetry in the excitation location and boundary conditions, the superposition of incident and reflected waves does not result in perfect cancellation. Instead, a net momentum flux is established, producing a mean flow. Analogies were drawn with arterial hemodynamics, where forward and backward pressure waves superimpose to form characteristic nodes and antinodes, critically influencing local stresses and flow patterns.

Resonance was identified as a key operating regime: when the excitation frequency approaches the natural frequencies of the coupled fluid–structure system, energy transfer is maximized, oscillation amplitudes increase, and efficient pulsatile flow can be sustained with reduced actuation effort.

2.4 High-Fidelity Modeling and Contact Mechanics

From a numerical standpoint, the system poses severe challenges. High-fidelity finite element models must simultaneously resolve large-strain structural mechanics, incompressible fluid flow, strong FSI coupling, and transient contact between the pinching mechanism and the deformable tube.

Contact mechanics was unanimously identified as the dominant computational bottleneck. Resolving contact constraints at every time step leads to extremely stiff nonlinear systems and prohibitive computational costs, rendering full-order simulations unsuitable for parametric studies, optimization loops, or control synthesis. Prior experience with industrial forming processes was cited to illustrate the intrinsic difficulty and poor scalability of brute-force contact resolution.

2.5 Reduced-Order Modeling for Design and Control

To overcome these limitations, the discussion converged on a projection-based reduced-order modeling strategy tailored to contact-dominated FSI problems. The key idea is to confine the computationally expensive resolution of contact and large deformation patterns to an offline stage, where representative operating scenarios are simulated using full-order models.

From these simulations, dominant deformation modes, contact states, and stress patterns are extracted and embedded into a reduced basis. During online evaluations, the governing equations are projected onto this low-dimensional space, allowing stresses, strains, and flow quantities to be evaluated only at a small subset of representative elements or integration points. Typical reductions from hundreds of thousands of degrees of freedom to a few tens of active evaluations were discussed.

It was explicitly stressed that this approach remains physics-based and interpretable, in contrast to purely data-driven regressions. The reduced models preserve explicit dependence on physical parameters and are therefore suitable for design optimization, uncertainty analysis, and real-time control.

2.6 Experimental Evidence and Implications for ECMO

Experimental setups featuring transparent compliant membranes were presented as a means of directly observing deformation and flow patterns. These experiments demonstrate that a constant inflow can be converted into a pulsatile, time-dependent outflow solely through structural compliance, without any active flow modulation.

This observation was highlighted as particularly relevant for ECMO applications, where standard rotary pumps deliver nearly steady flow. Introducing compliant, wave-driven elements could restore physiologically relevant pulsatility without increasing mechanical complexity.

High-speed imaging revealed complex, quasi-periodic vibration patterns, potentially involving multiple interacting modes. While the precise excitation and mode selection mechanisms remain under investigation, it was agreed that their characterization is essential for achieving stable, efficient, and controllable pump operation.

In summary, the discussion established a coherent technical roadmap in which high-fidelity FSI modeling, reduced-order representations, and targeted experiments jointly contribute to the understanding and exploitation of Liebau-based pumping mechanisms for extracorporeal circulatory support.

3 Laboratory

3.1 Laboratory Setup for Liebau-Based Pumping Experiments

3.2 Integration of the Pumping Module into an Extended Hydraulic Loop

Figure 2 illustrates the system-level experimental configuration in which the Liebau-based pumping module is embedded within an extended hydraulic loop. In contrast to isolated component testing, this arrangement allows the interaction between the local actuation mechanism and the global hydraulic environment to be investigated.

The extended tubing network introduces distributed inertial, resistive, and compliant effects that play a central role in wave propagation and reflection phenomena. These non-local effects are essential for the operation of Liebau-type pumps, whose performance depends not only on the local pinching action but also on the timing and superposition of pressure and deformation waves traveling along the circuit. By modifying tube lengths, reservoir positions, or downstream

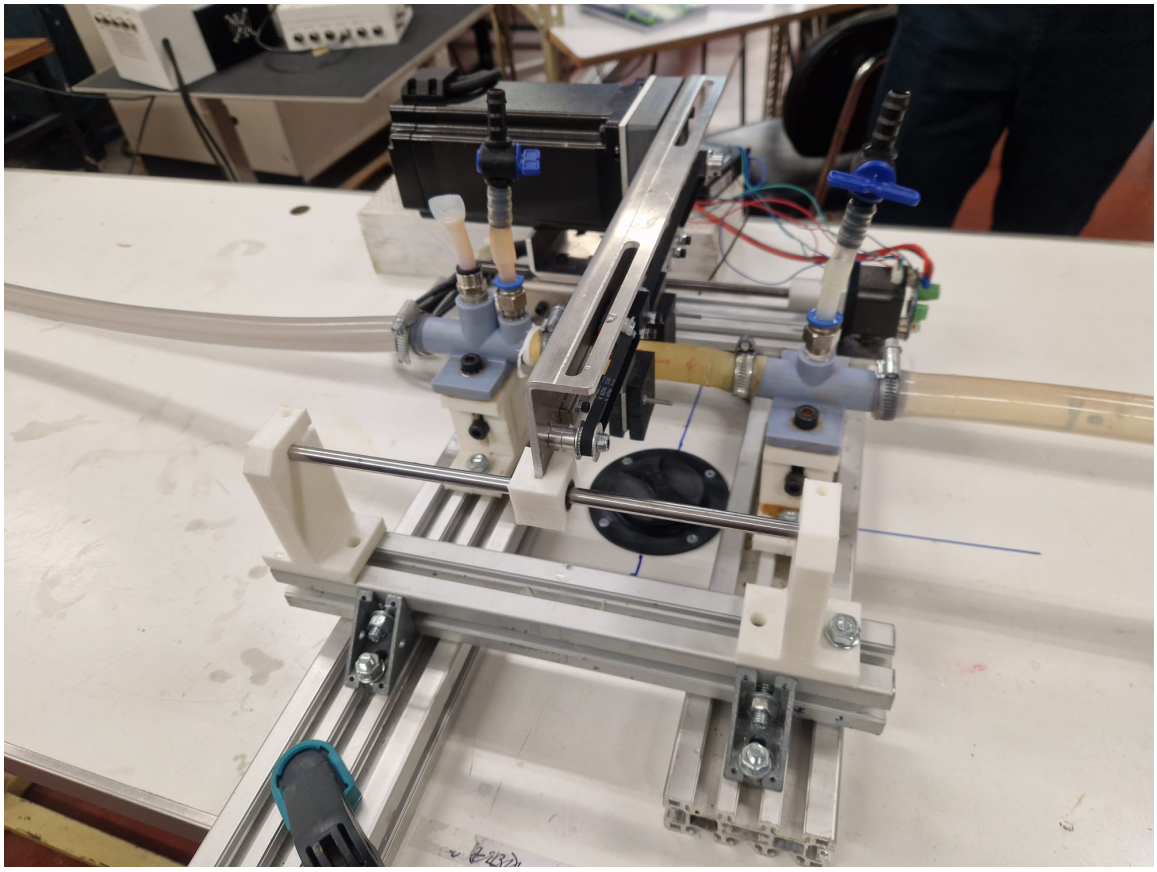

Figure 1: Laboratory-scale experimental setup used to investigate wave-driven flow generation in a compliant tube. A flexible transparent conduit is mounted within a rigid frame and subjected to localized periodic compression by a motor-driven linear actuator. The actuation mechanism applies controlled deformation to the tube without valves or rotating components, enabling the study of Liebau-type pumping effects. Rigid upstream and downstream sections, together with inline connectors and side ports, allow for well-defined boundary conditions and the integration of pressure or flow measurements. The transparent tube provides direct optical access for deformation and flow visualization.

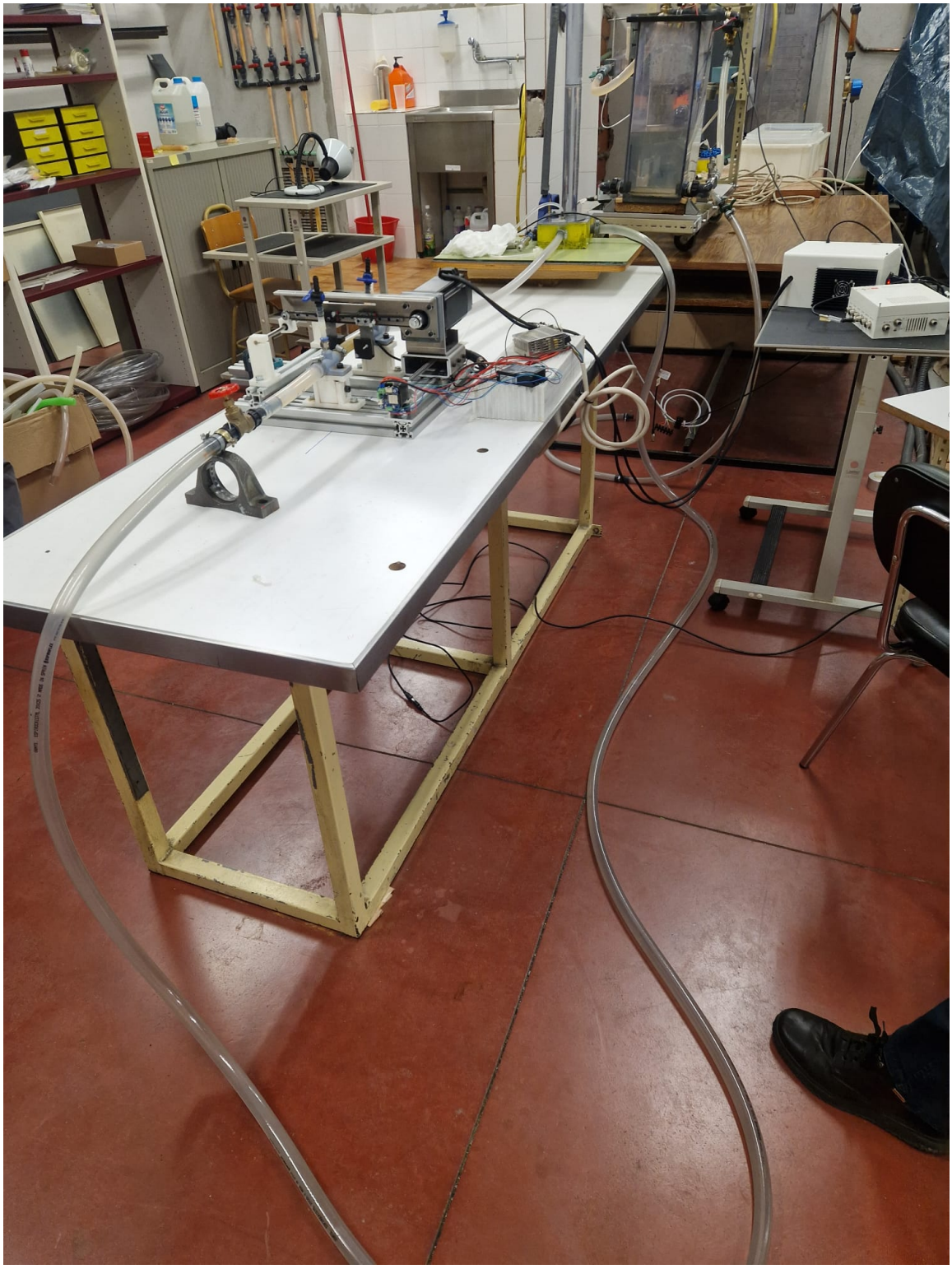

Figure 2: Extended laboratory configuration showing the integration of the Liebau-based pumping module within a larger hydraulic circuit. The compliant-tube test bench is connected via flexible tubing to auxiliary elements of the experimental loop, enabling controlled recirculation of water and the study of wave-driven pumping under non-local hydraulic boundary conditions.

impedance, the setup enables systematic exploration of resonance conditions and boundary-induced asymmetries.

From an experimental standpoint, the configuration supports continuous operation with water as the working fluid, while allowing the integration of pressure sensors and other measurement devices along the circuit. This makes it possible to quantify transient pressure fields, flow pulsatility, and phase relationships between actuation and system response.

From a modeling perspective, the setup provides a crucial link between high-fidelity local fluid–structure interaction simulations of the pinching region and reduced-order or lumped-parameter representations of the surrounding circuit. Experimental data obtained from this configuration are particularly relevant for validating boundary-condition models, calibrating wave-propagation parameters, and assessing the robustness of reduced-order models intended for design optimization and control applications.

Overall, the extended loop shown in Figure 2 represents an intermediate experimental scale between isolated benchtop demonstrations and full ECMO emulator integration, enabling progressive increases in system complexity while maintaining experimental control and interpretability.

3.3 Hydraulic Conditioning and Boundary-Condition Module

Figure 3 shows the hydraulic conditioning module employed to define and control the boundary conditions of the experimental loop. A vertical transparent column filled with water is used to impose a prescribed hydrostatic pressure level, allowing static head variations to be introduced in a simple and reproducible manner. The transparency of the column further enables direct visual inspection of fluid levels and transient effects.

The conditioning module is hydraulically connected to the rest of the experimental circuit, including the compliant-tube pumping section shown in Figures **??** and 2. By adjusting the fluid level in the vertical column and the configuration of the connected tubing, the effective downstream or upstream impedance of the system can be modified. This is particularly relevant for wave-driven pumping mechanisms, whose performance is strongly influenced by boundary-induced wave reflections and pressure offsets.

From a physical standpoint, the module allows decoupling of local actuation effects from global pressure constraints, making it possible to study how static pressure levels interact with dynamically generated pressure and deformation waves. This is essential for identifying operating regimes in which net flow generation is enhanced or suppressed due to constructive or destructive wave interference.

From a modeling and validation perspective, the hydraulic conditioning setup provides well-defined boundary conditions that can be directly incorporated into numerical simulations and reduced-order models. The controlled hydrostatic head serves as a reference pressure level for calibrating pressure measurements and validating the consistency between experimental observations and system-level models of the coupled fluid–structure dynamics.

Overall, the boundary-condition module shown in Figure 3 plays a central role in bridging localized compliant-tube experiments and more realistic extracorporeal circulation scenarios, supporting systematic investigation of resonance, impedance matching, and robustness of the

Figure 3: Hydraulic conditioning module used to impose and regulate boundary conditions in the experimental circuit. The setup includes a vertical transparent column providing a controllable hydrostatic head, together with reservoirs, tubing, and auxiliary components for flow conditioning and pressure stabilization. This module is connected to the compliant-tube pumping system to investigate the influence of static head, impedance, and wave reflection on the performance of the Liebau-based pump.

Liebau-based pumping concept.

3.4 Detail of the Actuation, Pinching Mechanism, and Instrumentation

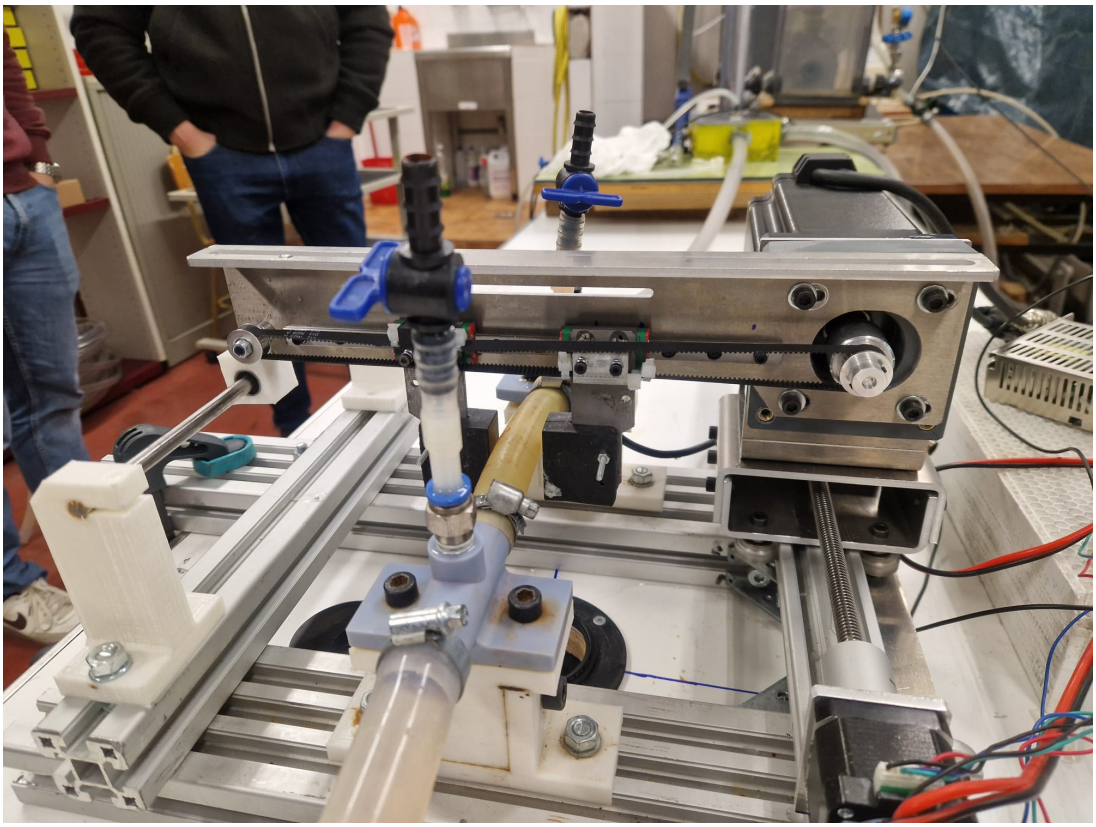

Figure 4: Detail view of the Liebau-based pumping module highlighting the linear actuation system, pinching mechanism, and local instrumentation. A motor-driven carriage operating under prescribed displacement control translates along linear guides and applies localized compression to the compliant tube. Inline fittings with valves and side ports allow for pressure measurements and flow control in the immediate vicinity of the actuation region.

Figure 4 provides a close-up view of the core mechanical and hydraulic components responsible for flow generation in the Liebau-based pumping setup. The linear actuation system consists of a motorized stage mounted on precision guides, ensuring repeatable and well-defined kinematics. The actuator imposes a prescribed displacement on the pinching element, thereby decoupling the imposed motion from the resulting fluid–structure interaction forces.

The pinching mechanism locally compresses the compliant tube, inducing transient deformation of the tube wall and generating pressure waves that propagate along the fluid-filled conduit. The geometric simplicity of the pinching element, together with the rigid support structure, facilitates accurate modeling of contact conditions and deformation patterns in numerical simulations.

Instrumentation elements are integrated directly upstream and downstream of the actuation zone. Inline connectors equipped with valves and side ports allow the placement of pressure sensors and the regulation of local flow conditions. This configuration enables high temporal resolution measurements of pressure fluctuations associated with wave generation, reflection,

and interference, which are central to the operation of Liebau-type pumps.

From a modeling perspective, the localized nature of the actuation and contact region shown in Figure 4 motivates the use of high-fidelity fluid–structure interaction models for this subdomain, combined with reduced-order or lumped representations for the remainder of the hydraulic circuit. The clear separation between the actuation mechanism, the compliant tube, and the surrounding rigid supports provides a well-defined basis for both full-order simulations and projection-based model reduction strategies aimed at control and design optimization.

3.5 Fabrication of the Compliant Tube

Figure 5: Setup used for the fabrication of the compliant tube employed in the Liebau-based pumping experiments. A vertical cylindrical mold is filled with a liquid latex-based compound, allowing controlled deposition and curing of a thin-walled elastic tube. The process enables reproducible fabrication of compliant conduits with tailored geometric and mechanical properties.

Figure 5 shows the laboratory setup used to fabricate the compliant tubes employed in the experimental pumping system. The device consists of a vertical cylindrical mold partially filled with a liquid latex-based material, into which a mandrel or inner core is inserted to define the internal diameter of the tube. By controlling immersion depth, residence time, and material properties, a thin elastic layer is deposited along the mold.

The vertical configuration promotes uniform thickness and minimizes gravitational asymme-

tries during curing. Once solidified, the compliant tube can be extracted from the mold and integrated into the experimental circuit shown in Figures **??**–**??**. This fabrication approach allows the mechanical properties of the tube, such as wall thickness, stiffness, and compliance, to be tuned according to the requirements of the pumping experiments.

From a physical standpoint, the ability to manufacture compliant tubes in-house is critical for systematic investigation of Liebau-based pumping mechanisms, as wave propagation, resonance, and flow generation are highly sensitive to wall elasticity and geometric parameters. From a modeling perspective, the controlled fabrication process facilitates consistency between experimental specimens and the material models used in numerical simulations, including fluid–structure interaction and reduced-order representations.

Overall, the tube fabrication setup shown in Figure 5 constitutes a key enabling element of the experimental campaign, ensuring repeatability and parametric control over one of the most influential components of the pumping system.

3.6 Ultrasonic Flow Measurement Device

Figure 6: Clamp-on ultrasonic flow sensor (Sonotec SONOFLOW) installed on the compliant tube for non-invasive flow rate measurements. The sensor operates without direct contact with the fluid, preserving the integrity and compliance of the elastic tube while enabling time-resolved flow monitoring.

Figure 6 shows the ultrasonic flow measurement device used to quantify flow rates in the experimental circuit. The sensor is clamped externally onto the compliant tube and operates based on ultrasonic transit-time principles, allowing non-invasive measurement of volumetric

flow without introducing additional pressure losses or mechanical disturbances.

The use of a clamp-on sensor is particularly advantageous in the context of Liebau-based pumping experiments, where flow rates are typically unsteady and strongly coupled to wall deformation. By avoiding direct contact with the fluid, the sensor preserves the mechanical properties of the compliant tube and prevents alterations to wave propagation characteristics that could arise from intrusive measurement devices.

From an experimental standpoint, the flow sensor provides time-resolved measurements suitable for capturing pulsatile flow patterns induced by wave-driven pumping. These data are essential for correlating actuation parameters, pressure fluctuations, and resulting net flow generation.

From a modeling and validation perspective, the measured flow signals serve as key reference quantities for validating numerical simulations and reduced-order models of the coupled fluid–structure system. The non-invasive nature of the sensor ensures that experimental observations can be directly compared with idealized model predictions, without the need to account for measurement-induced perturbations.

Overall, the ultrasonic flow sensor shown in Figure 6 constitutes a critical instrumentation component for assessing the performance, efficiency, and stability of the Liebau-based pumping mechanism under controlled experimental conditions.

3.7 Dynamic Response of the Compliant Tube Under Periodic Actuation (Video)

A video recording of the experiment is provided to illustrate the dynamic behavior of the compliant tube under periodic external actuation. The video can be accessed at the following link:

`FIGS/VID1.mp4`

The video shows the time-dependent deformation of the elastic tube and the associated wave propagation induced by the prescribed-displacement actuation mechanism. The motion highlights the generation of pressure and wall-deformation waves that travel along the compliant conduit, interact with boundary conditions, and give rise to pulsatile flow without the use of valves or rotating components.

Qualitatively, the video evidences the strongly coupled nature of the fluid–structure interaction: localized tube compression produces global deformation patterns, with visible phase shifts and amplitude modulation along the tube length. Depending on the actuation parameters and hydraulic boundary conditions, the response exhibits quasi-periodic behavior, suggesting the superposition of multiple vibration modes rather than a single harmonic response.

This visual evidence is particularly valuable for interpreting experimental measurements and validating numerical models. The observed deformation patterns and wave dynamics provide direct insight into resonance phenomena, wave reflection, and impedance effects, which are central to the operation of Liebau-based pumping mechanisms. The video therefore complements the pressure and flow measurements discussed previously and supports the development and validation of high-fidelity and reduced-order models of the coupled system.

3.8 Wave Propagation and Flow Asymmetry in the Extended Circuit (Video)

A second video recording is provided to illustrate the response of the compliant tube and the surrounding hydraulic circuit under periodic actuation. The video can be accessed at the following link:

`FIGS/VID2.mp4`

The video focuses on the propagation of deformation and pressure waves beyond the immediate actuation region, highlighting their interaction with the extended hydraulic loop. In contrast to the localized view presented in the previous video, this recording emphasizes the non-local effects induced by wave reflection, impedance mismatches, and circuit geometry.

The observed motion reveals a clear asymmetry in the system response, consistent with the Liebau pumping principle. Waves generated at the pinching location propagate along the compliant tube, reflect at boundaries, and return with phase shifts that prevent symmetric cancellation. The resulting spatio-temporal superposition produces a net directional transport of fluid, despite the absence of valves or rotating elements.

From a qualitative standpoint, the video illustrates how global boundary conditions strongly influence the effectiveness of wave-driven pumping. Variations in tube length, reservoir position, and hydraulic head affect the timing and amplitude of reflected waves, thereby modifying the resulting flow pulsatility and net flow rate. These effects are difficult to infer from local measurements alone and underscore the importance of system-level visualization.

This video provides essential experimental evidence supporting the modeling strategy adopted in the project, namely the combination of detailed local fluid–structure interaction models with reduced-order or lumped representations of the surrounding circuit. The visualized wave dynamics offer direct guidance for the selection of boundary conditions, resonance tuning, and validation of numerical predictions.

4 Post-Laboratory Technical Meeting: Strategy and Future Work

This section summarizes the technical discussion held after the laboratory visit, whose purpose was to assess the experimental observations and define a coherent strategy for future experimental and numerical work. The meeting did not involve new experiments, but rather focused on interpreting the laboratory setup and observations in light of the project objectives, and on establishing a structured roadmap for subsequent investigations.

4.1 Role of the Laboratory Observations

The laboratory visit provided direct insight into the physical realization of the compliant-tube pumping concept, including actuation mechanisms, boundary-condition control, instrumentation, and tube fabrication procedures. These observations served as a concrete reference for identifying both opportunities and limitations of the current setup.

Rather than attempting to immediately reproduce the full experimental configuration numerically, the discussion emphasized the need to decompose the problem into progressively more

complex stages, each targeting a specific physical mechanism. This staged approach was considered essential to ensure numerical robustness, interpretability, and meaningful validation.

4.2 Definition of a Phased Research Strategy

A multi-stage strategy was proposed to guide future work, with increasing levels of physical and numerical complexity.

Stage 1: Structural-Only Analysis. The first proposed stage consists of studying the compliant tube under prescribed displacement actuation in the absence of fluid effects (air inside and outside the tube). The objective of this stage is not to model pumping, but to understand large-deformation structural behavior, including contact and self-contact, under controlled kinematic loading.

This stage is intended to:

- establish a reliable structural finite element model of the tube,
- identify deformation patterns and contact regimes,
- generate reference data suitable for subsequent reduced-order modeling of the solid.

Stage 2: Local Fluid–Structure Interaction. The second stage introduces water inside the tube, while keeping hydraulic boundary conditions as simple and local as possible. The tube is conceptually connected to nearby reservoirs imposing approximately constant pressure conditions, thereby limiting inertial and wave-propagation effects associated with long rigid segments.

This stage targets validation of fully coupled 3D fluid–structure interaction (FSI) models under conditions where the dominant physics arises from the interaction between wall deformation and fluid motion, rather than from global circuit dynamics.

Stage 3: System-Level Wave and Inertia Effects. In the third stage, the full hydraulic circuit is considered, including extended rigid tubes upstream and downstream of the compliant section. In this regime, wave propagation, reflection, and inertial effects become central to the pumping mechanism.

It was explicitly noted that although the numerical formulation may appear similar to Stage 2, the underlying physical mechanism differs substantially. This distinction motivates treating the stages separately rather than attempting to address all effects simultaneously.

4.3 Initial Stress, Pretension, and Reference Configuration

A key outcome of the discussion was the recognition that simulations starting from a stress-free configuration are neither numerically robust nor physically representative for highly compliant tubes. Instead, a well-defined initial equilibrium state must be established prior to dynamic actuation.

The agreed strategy consists of:

- introducing pretension or gradual pressurization to reach a stable equilibrium configuration,
- using this configuration as the reference state for subsequent transient simulations,
- avoiding abrupt loading that could trigger non-physical instabilities.

This approach was explicitly connected to standard practices in arterial mechanics and soft-tissue modeling.

4.4 Contact Modeling and Actuator Design Considerations

The meeting highlighted contact modeling as one of the principal numerical challenges for future work. Sharp or flat contact geometries were identified as a source of severe numerical difficulties, motivating the use of smooth, rounded actuator profiles in both experiments and simulations.

From a planning perspective, it was agreed that actuator geometry should be treated as a design variable, not only for experimental convenience but also to facilitate numerical convergence and reduced-order modeling.

4.5 Coupling High-Fidelity Models and Reduced-Order Representations

The discussion clarified the intended relationship between experiments, full-order simulations, and reduced-order models (ROMs). Experimental data are primarily intended to validate high-fidelity 3D models, including contact and large-deformation effects. These validated models then serve as the basis for constructing ROMs.

ROMs are envisioned as:

- acceleration tools for parametric studies,
- enablers for system-level simulations,
- potential components of future control strategies.

Direct validation of ROMs against experiments was not considered necessary at this stage; instead, ROMs should be benchmarked against their corresponding full-order models.

4.6 System-Level Modeling Perspective

For configurations involving extended rigid segments, the need for hybrid modeling approaches was emphasized. In such cases, 3D FSI models of the compliant tube should be coupled with 1D or 0D representations of the surrounding hydraulic circuit.

This hybrid perspective was identified as essential for achieving computationally feasible simulations while preserving the physical mechanisms governing wave-driven pumping.

4.7 Outcome of the Meeting

The post-laboratory meeting resulted in a shared understanding of the problem decomposition required to address Liebau-based pumping in a rigorous and scalable manner. The staged strategy aligns experimental feasibility with numerical robustness and provides a clear roadmap for future collaborative work between experimental and computational teams.

4.8 Closing Discussion: Material Properties, Calibration, and Immediate Next Steps

During the final part of the meeting, the discussion focused on practical aspects related to material modeling, numerical feasibility, and the immediate actions required to initiate simulations and reduced-order model development.

It was noted that a three-dimensional modeling effort would be required to properly address contact phenomena, even if simplified geometries such as spherical or circular pinching elements were initially considered. In this context, attention turned to the mechanical properties of the compliant tube material and the availability of reliable reference data.

The participants mentioned the existence of a manuscript developed at the University of Valladolid that, while not yet formally published, has been used internally to calibrate material properties of flexible tubes. It was clarified that this work is available as a preprint in the University of Valladolid repository and has already served as a basis for previous simulations. The material parameters employed in that work were confirmed to be consistent with those commonly used in commercial finite element software, such as ANSYS, alleviating concerns regarding compatibility.

Based on this, it was agreed that these calibrated material properties could be directly adopted as a starting point for the present project, enabling the team to proceed without delay to preliminary three-dimensional simulations. In particular, the calibrated properties were considered sufficient for initiating the first numerical case, corresponding to a simplified configuration with imposed displacement and without explicit fluid modeling.

In parallel, it was suggested that reduced-order modeling activities could begin independently of the full contact problem, by initially neglecting contact and focusing on displacement-driven structural response. This decoupled approach was seen as a way to make early progress on reduced-order formulations while high-fidelity simulations are being prepared.

The discussion also addressed experimental and post-processing aspects, notably the acquisition and analysis of high-speed camera recordings. Concerns were raised regarding temporal synchronization and spatial scaling of the recorded images, based on issues encountered in previous experiments. It was emphasized that careful verification of image scaling and reference dimensions would be required to ensure reliable segmentation and quantitative interpretation of deformation patterns. The possibility of leveraging existing image-processing workflows and segmentation tools was mentioned as a parallel task that could proceed while simulations are being set up.

Finally, it was agreed that, with the availability of calibrated material data and a clear initial numerical target, the project could formally transition from exploratory discussion to active implementation, with experimental refinement and numerical modeling advancing in parallel.

4.9 Reference Framework for Compliant Tube Material Characterization

The material calibration strategy discussed above relies on prior work carried out at the University of Valladolid on the characterization of flexible tube materials under pressurization. This work proposes a simple and robust methodology to extract constitutive parameters of compliant tubes by analyzing their static, axisymmetric deformation under a prescribed transmural

pressure.

In this framework, a thin-walled elastic tube is subjected to internal pressurization, and its resulting deformation is measured and compared against numerical simulations based on hyperelastic material models. By matching experimental and numerical responses, effective material parameters—such as shear modulus or equivalent stiffness measures—are identified. The approach is particularly well suited for highly compliant materials undergoing large deformations, for which standard uniaxial tests may be insufficient or impractical.

The methodology has been validated experimentally and numerically and provides direct access to quantities required for finite element simulations, including displacement fields, stress distributions, and deformation gradients. Although the reference configuration is axisymmetric and static, the resulting material parameters are directly applicable to three-dimensional simulations involving dynamic loading, contact, and fluid–structure interaction.

This reference work constitutes a key enabling element for the present project, as it provides a consistent and experimentally validated basis for defining the mechanical properties of the compliant tubes used in the Liebau-based pumping experiments. The calibrated material models serve as the foundation for both high-fidelity finite element simulations and subsequent reduced-order model construction [5].

5 Visit to Hospital Clínico de Valladolid: Clinical Context and Reference Pump Technology

Following the laboratory activities and the subsequent technical meeting, the project team visited the Hospital Clínico Universitario de Valladolid to discuss clinical requirements and constraints associated with extracorporeal circulation systems. The meeting was held with a clinician (Alex), originally from Belarus, who is also involved in teaching activities related to cardiovascular support technologies. The purpose of the visit was to clarify clinically relevant operating conditions—such as pressure levels, flow rates, and usage constraints—and to examine the rotary pump technology currently employed in practice, which constitutes the reference system targeted for replacement in the present project.

5.1 Reference Centrifugal Pumps Used in Current ECMO Practice

The clinical data and operating ranges discussed during the meeting at Hospital Clínico Universitario de Valladolid implicitly refer to current-generation centrifugal blood pumps used in ECMO systems. These devices constitute the technological baseline against which any alternative pumping concept must be evaluated.

Representative examples of such pumps include the Maquet Rotaflow system, the LivaNova (formerly Sorin) CentriMag pump, the Medtronic Bio-Console platforms, and the Terumo Capiox series [2, 3, 4, 6]. Although differing in detailed design, these pumps share common operating principles and performance envelopes.

Centrifugal ECMO pumps typically operate by imparting kinetic energy to the blood via a rapidly rotating impeller, generating a pressure rise sufficient to drive flow through the extracorporeal circuit and patient vasculature. Reported operating characteristics include:

- nominal flow rates between 2 and 7 L/min,
- achievable pressure rises on the order of 200–500 mmHg,
- rotational speeds of several thousand revolutions per minute,
- continuous, non-pulsatile flow delivery.

These devices are widely adopted due to their compactness, reliability, and ease of integration with oxygenators and monitoring systems. However, they are also associated with known limitations, including sustained exposure of blood to high shear stresses, the formation of recirculation zones within the pump housing and connectors, and an increased risk of hemolysis and thrombogenesis under low-flow or high-resistance conditions [1].

Throughout the clinical discussion summarized in the following subsections, the pressure thresholds, flow ranges, alarm limits, and failure modes described by the clinician should be interpreted as representative of these centrifugal ECMO pump systems. -

5.2 Overview of the Reference Rotary Pump Technology

During the discussion, the clinician presented a compact rotary pump unit currently used in extracorporeal circulation procedures, including ECMO. The device was shown directly during the meeting and is illustrated from multiple viewpoints in Figures 7–11.

The device corresponds to a centrifugal rotary blood pump, a class of pumps widely used in ECMO and other extracorporeal support systems. In such devices, blood enters axially through a central inlet, is accelerated radially by a rotating impeller, and exits tangentially through an outlet port. Flow generation is achieved through continuous rotation rather than volumetric displacement, resulting in a non-pulsatile or weakly pulsatile flow.

Typical operating characteristics for pumps of this class include:

- volumetric flow rates on the order of 3–6 L/min for adult ECMO support,
- pressure rises up to approximately 300–500 mmHg depending on rotational speed and circuit resistance,
- rotational speeds in the range of several thousand revolutions per minute,
- compact, disposable pump heads designed to minimize blood residence time.

5.3 Clinical Motivation for Alternative Pumping Concepts

The rotary pump presented during the meeting serves as the current clinical standard against which alternative pumping concepts must be evaluated. While centrifugal pumps offer reliable flow control and compact integration, they are associated with well-known limitations, including continuous exposure of blood to high shear rates, non-physiological flow patterns, and potential hemolysis and thrombogenicity under certain operating conditions.

The discussion at the Hospital Clínico emphasized that any alternative technology—such as the valveless Liebau-based pump investigated in this project—must ultimately meet or exceed

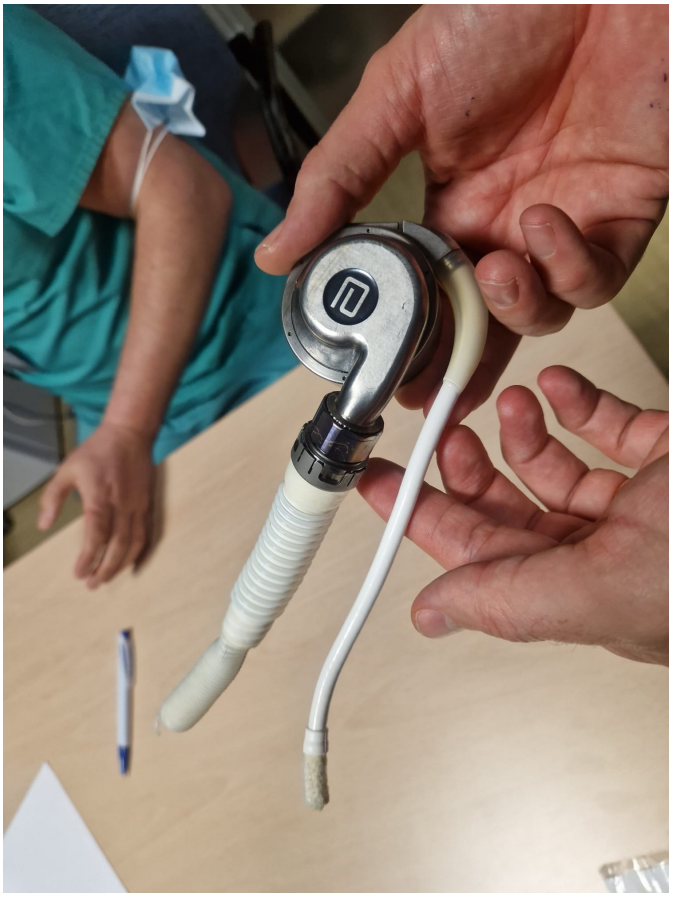

Figure 7: Reference rotary pump used in extracorporeal circulation, shown with attached inflow and outflow tubing. The compact housing integrates the rotating impeller and blood-contacting components.

Figure 8: Side view of the rotary pump highlighting the curved flow path and connector geometry designed to minimize sharp turns and flow separation.

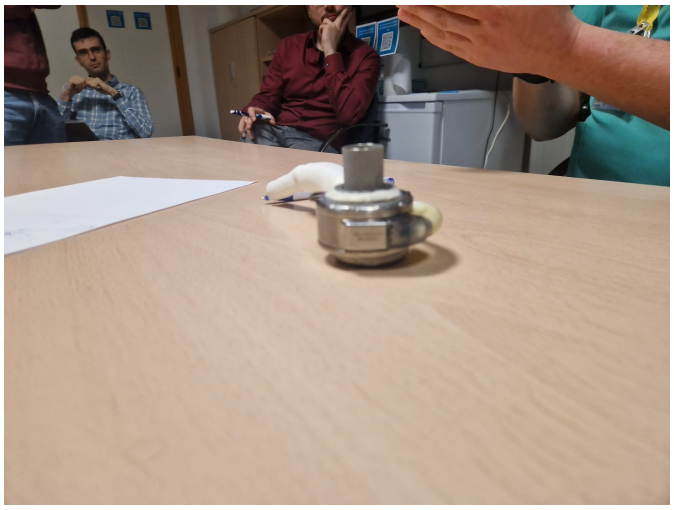

Figure 9: Top view of the rotary pump housing, showing the central inlet region and the circular symmetry typical of centrifugal blood pumps.

Figure 10: Detail of the pump inlet section, illustrating the axial entry of the fluid toward the impeller region.

Figure 11: Alternative view of the rotary pump unit presented during the clinical meeting, emphasizing its compact size and integrated tubing connections.

the performance envelope of these rotary devices in terms of delivered flow rate, pressure capability, and operational robustness, while offering tangible benefits in terms of blood compatibility and physiological flow characteristics.

The reference pump shown in Figures 7–11 therefore provides a concrete clinical benchmark. Its geometry, operating range, and integration constraints define the baseline requirements that guide both experimental design and numerical modeling efforts in the subsequent phases of the project.

5.4 Clinical Requirements Discussion at Hospital Clínico: Flow, Pressure, and Circuit Constraints

This subsection summarizes the first part of the clinical discussion held at Hospital Clínico Universitario de Valladolid. The questions were primarily posed by José Sierra (UVA), focusing on operational envelopes and specification ranges required to start engineering work on a valveless, Liebau-based pumping concept. The answers were provided by Alex (clinician), who described typical ECMO settings, alarm thresholds, and practical failure modes observed in current rotary (centrifugal) pumping systems.

5.4.1 Motivation: establishing a realistic operating envelope (José Sierra)

José Sierra opened by clarifying that the project team already has a proof-of-concept based on periodic compression of a compliant tube (valveless actuation), but that the initial technical need is to anchor the development to clinically meaningful ranges: target flow rates, admissible pressure head, and typical suction/negative-pressure values at the venous drainage side. A key request was whether a commercial datasheet or standard specification sheet could be used as a baseline reference (pressure rise, nominal flow, etc.), to avoid designing around unrealistic requirements.

Alex noted that a single "universal" operating range is difficult to state abstractly without reference to circuit configuration and patient condition, but then provided representative ranges used in practice and the corresponding alarm thresholds.

5.4.2 Typical adult flow rates and practical limits (Alex)

Alex stated that typical adult ECMO flows are in the range of approximately 2–7 L/min, with the practical observation that the upper end (around 7 L/min) is rarely used because it requires an excellent venous preload (adequate venous return) to avoid collapse of the inlet line. In routine clinical settings, flows around 3–4 L/min were described as common operating points.

The clinician emphasized that high flow demands can drive the system toward more negative inlet pressures (venous drainage suction), potentially causing inlet-line chatter/collapse and unstable flow delivery.

5.4.3 Venous drainage (preload) pressures and suction alarms

A central part of the discussion concerned the pressure levels at the venous drainage side (inlet to the extracorporeal circuit). Alex explained that:

- When expressed relative to central venous pressure (CVP), "physiological" positive values are around 10–14 mmHg, but extracorporeal drainage often operates at negative pressures.
- For typical clinical flow delivery of roughly 3–4 L/min, inlet pressures on the order of -50 to -60 mmHg can occur in practice.
- An alarm threshold is commonly set at approximately -70 mmHg at the circuit inlet (venous drainage side). If the inlet pressure becomes more negative than this value, the system triggers an alarm.

The mechanism was described in practical terms: if the patient becomes hypovolemic (e.g., bleeding, diuresis) or if venous return is otherwise compromised, maintaining the same pump setting requires higher suction, which can collapse the inlet line and induce oscillations ("the inlet line starts to dance"). The pump-console protection algorithms may automatically reduce rotational speed to mitigate suction, which is clinically dangerous if it happens silently: the system can decrease speed progressively (to suppress suction signatures), and the operator may fail to notice a significant loss of delivered flow unless alarms and monitoring are interpreted correctly.

5.4.4 Pressure rise (head) produced by centrifugal pumps and arterial-side alarms

José Sierra then asked for the order of magnitude of the pressure head required to drive the full circuit (pump + oxygenator + patient).

Alex stated that centrifugal pumps in ECMO commonly generate pressure rises on the order of 200–300 mmHg, and that small devices can reach approximately 300 mmHg as a typical "top" range. The total pressure is distributed across the oxygenator and circuit losses, and the pressure delivered to the patient depends on cannula size and downstream resistance.

For the arterial return to the patient, Alex noted a positive-pressure alarm threshold around 400 mmHg (measured near the return cannula), while routine operation is commonly within $\sim$ 200–300 mmHg depending on the specific configuration and patient/cannula.

5.4.5 Cannula dimensions and geometric constraints of the circuit

To translate the discussion into design constraints, Alex provided representative cannula sizes:

- Arterial return cannula: typically 15–17 Fr (approximately 5–6 mm internal diameter scale, using the clinical convention 1 Fr $\approx$ 1/3 mm).
- Venous drainage cannula: commonly larger, around 21–24 Fr (drainage side), to reduce inlet losses and suction requirements.

In addition, the main tubing diameter in standard ECMO circuits was indicated as *three-eighths of an inch* (3/8"), and it was suggested that providing a short segment of tubing (and, when possible, a clean cannula sample no longer intended for use) would be valuable for the engineering team to reproduce the hydraulic impedance and connector geometry in laboratory prototypes.

5.4.6 Circuit priming volume and blood inventory outside the patient

Another explicit question from José Sierra concerned the amount of blood contained in the extracorporeal circuit.

Alex estimated that the oxygenator plus tubing volume is roughly on the order of ~ 0.5 L, and that, during decannulation/removal, clinicians typically recover about 500 mL (two blood bags) from the circuit. In practice, connecting a patient requires an additional priming volume of roughly another half liter scale, depending on the exact circuit/oxygenator configuration.

5.4.7 Low-flow regime, thrombosis risk, and hemolysis-driven coagulation (first mention)

Although more detailed thrombosis mechanisms were discussed later, Alex already highlighted (in this first part) two clinically important qualitative facts:

- Thrombotic risk in rotary pumps and oxygenators increases when flows are reduced, with a critical regime around 2 L/min and below.
- Hemolysis is not only a red-cell damage issue: the release of intracellular mediators (Alex explicitly mentioned ADP in this discussion) promotes platelet aggregation and can accelerate thrombus formation, especially in regions of recirculation and low washout.

5.4.8 Handwritten working notes and schematic recap

During the conversation, the team produced working sketches summarizing: (i) typical adult flow ranges, (ii) venous inlet suction thresholds (alarm around -70 mmHg), (iii) pressure rise across the pump and pressure at the return cannula, and (iv) representative tubing and cannula sizes (3/8" lines, return cannula around 15–17 Fr, drainage cannula larger around 21–24 Fr). These notes are provided for traceability in Figures 12–15.

5.4.9 Stage 2: Thrombosis, Hemolysis, and Flow-Related Failure Modes

Transition from flow specification to blood damage mechanisms (José Sierra). After establishing the typical operating envelope in terms of flow rates and pressures, José Sierra asked explicitly about the mechanisms by which current rotary pumps and extracorporeal circuits fail from a hemocompatibility standpoint. In particular, he asked whether thrombosis is primarily associated with low-flow operation, high shear, or geometric features of the pump and connectors, and how these effects manifest clinically.

Low-flow thrombosis and critical thresholds (Alex). Alex explained that thrombus formation in extracorporeal circuits is strongly correlated with low-flow regimes. In practice, flows below approximately 2 L/min are considered critical, and operation in this regime is avoided whenever possible. He stated that, statistically, fewer than 10% of patients develop clinically significant thrombosis in the pump itself, but that this risk increases markedly when flow drops below this threshold.

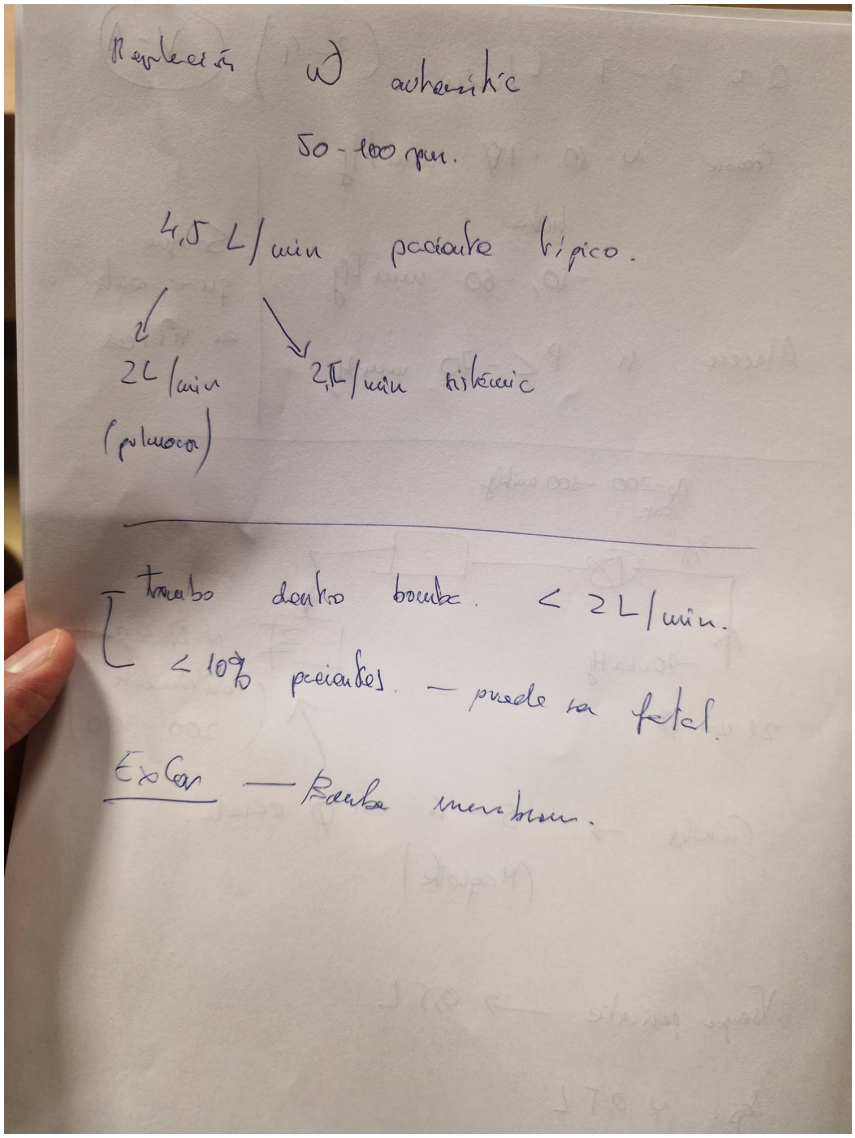

Figure 12: Handwritten note sheet (1): qualitative summary of adult flow levels and operational remarks recorded during the clinical discussion.

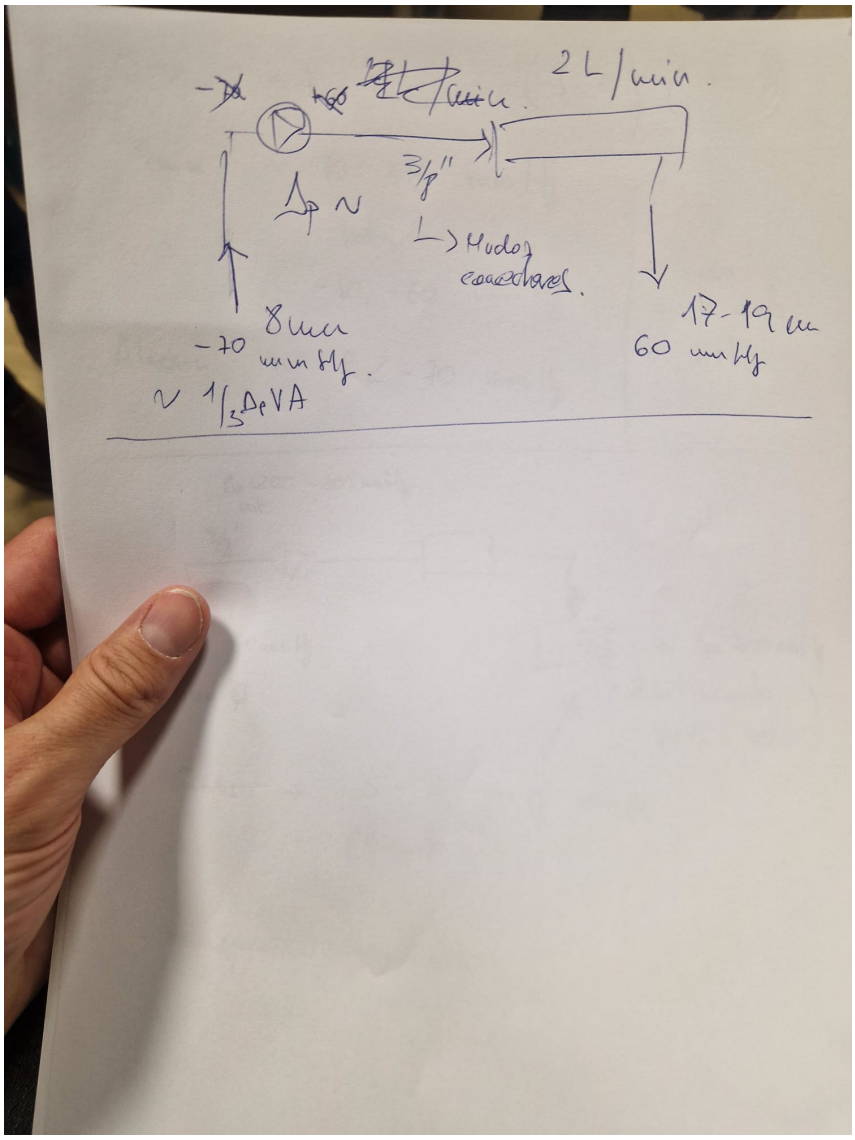

Figure 13: Handwritten note sheet (2): sketched circuit with indicative pressures (venous suction and arterial-side pressures) and tubing/cannula references.

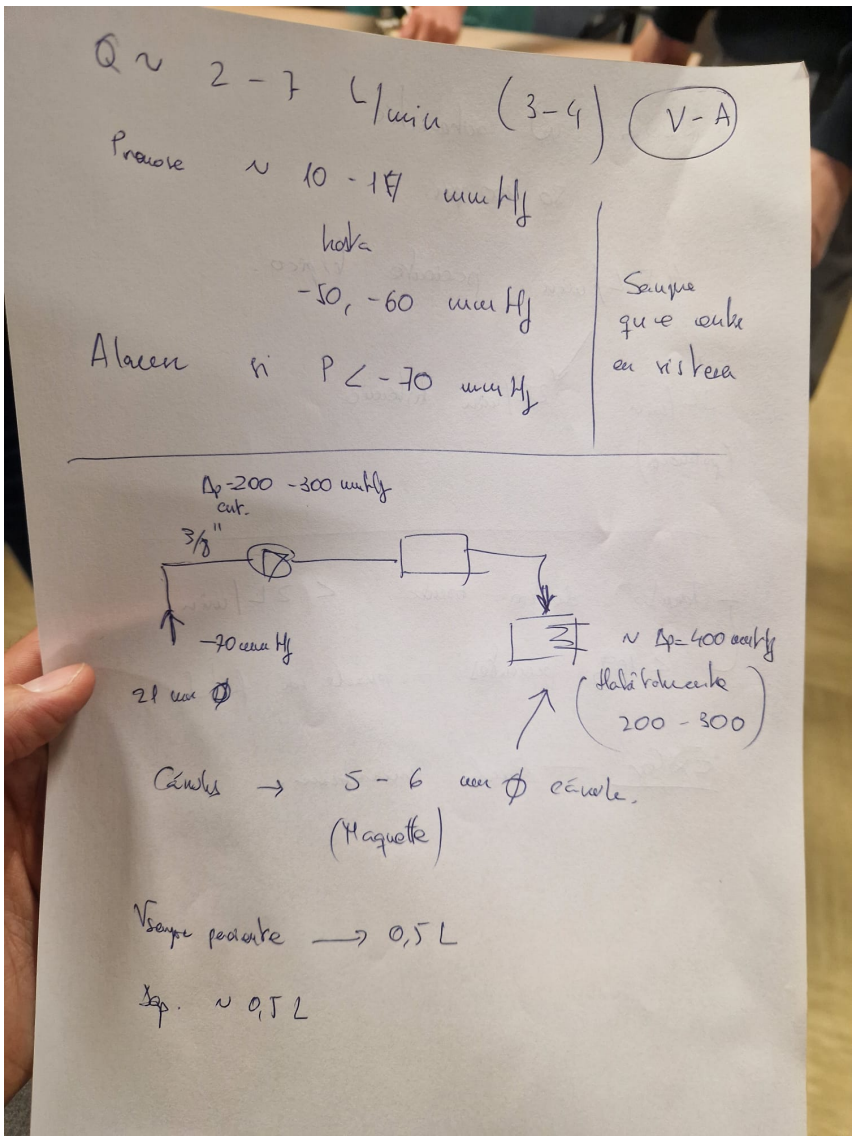

Figure 14: Handwritten note sheet (3): simplified flow/pressure diagram highlighting inlet suction levels and representative return pressures.

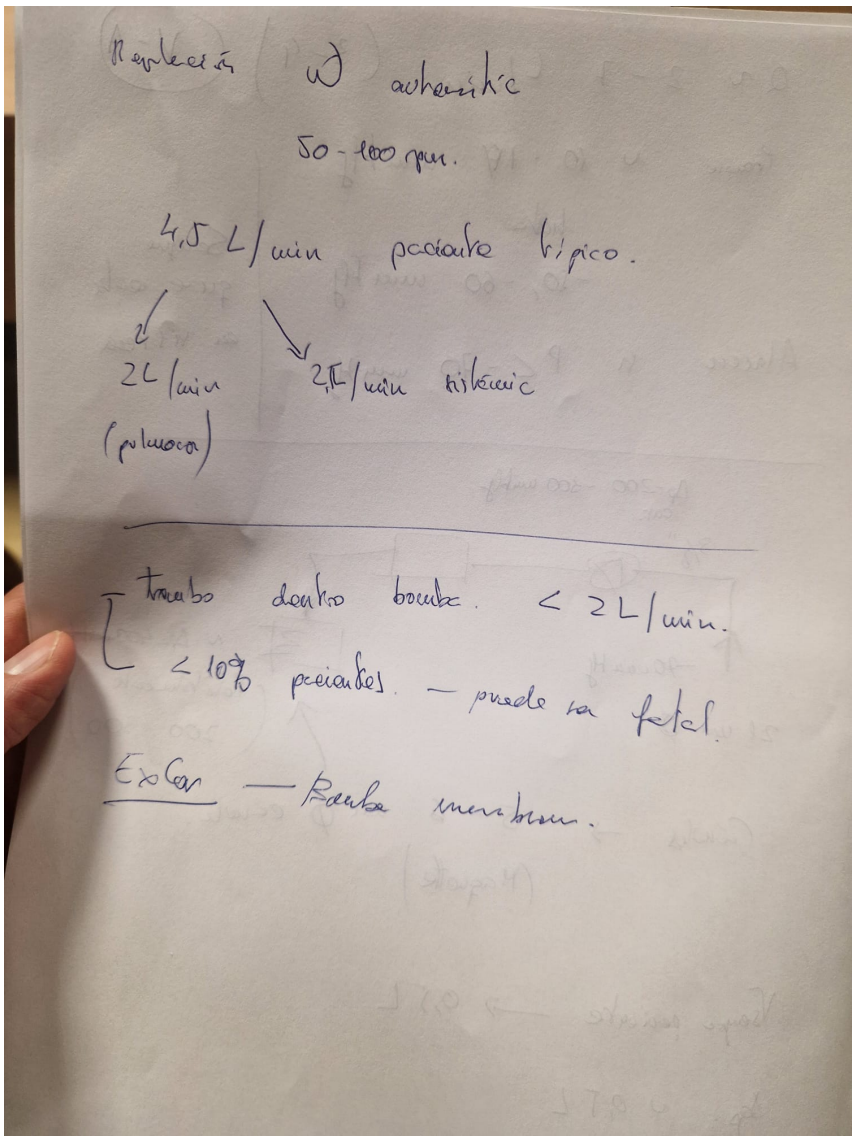

Figure 15: Handwritten note sheet (4): additional recorded numerical ranges and qualitative notes used to consolidate the engineering design envelope.

The underlying reason is not solely reduced bulk flow, but the appearance of stagnant or weakly washed regions within the pump housing and circuit components. These regions allow platelets and coagulation factors to remain in contact with artificial surfaces for extended residence times, favoring clot initiation.

Hemolysis as an indirect driver of coagulation. A key point emphasized by Alex was that hemolysis should not be viewed only as red blood cell damage. When red blood cells rupture, intracellular components are released into the plasma. Among these, adenosine diphosphate (ADP) plays a particularly important role as a potent platelet activator.

Thus, hemolysis indirectly promotes thrombosis by increasing platelet aggregation. In rotary pumps, hemolysis is driven by high shear regions near the impeller and by repeated exposure of blood to such regions during recirculation. Even if global flow appears adequate, local shear-induced hemolysis can therefore accelerate clot formation elsewhere in the circuit.

Geometric sources of recirculation and stagnation. Alex stressed that thrombus formation is not uniformly distributed throughout the circuit. Instead, it is strongly localized near geometric discontinuities, such as:

- connectors and reducers,
- abrupt diameter changes,
- sharp bends or elbows,
- regions immediately upstream or downstream of the pump housing.

In these areas, flow separation and recirculation zones arise even at nominal flow rates. These zones act as traps for activated platelets and fibrin, especially when combined with low overall flow or transient reductions in flow due to suction control.

This observation was highlighted as particularly relevant for engineering design: even if a pump delivers the correct mean flow and pressure, poor geometric transitions can dominate hemocompatibility outcomes.

Clinical illustration via patient imaging (Alex). At this point in the discussion, Alex showed an image on his mobile phone corresponding to a real patient case (the image itself is not reproduced here). He used this image to illustrate asymmetric perfusion patterns and clot-related complications observed in practice.

The intent of showing the image was not diagnostic detail, but rather to emphasize how flow distribution and circuit configuration can lead to clinically significant consequences that are not immediately obvious from pump console readouts alone. In particular, regions of poor washout and altered flow direction can persist even when global flow and pressure values appear acceptable.

Implications for alternative pumping concepts. Alex concluded this part of the discussion by noting that any alternative pumping technology aiming to replace centrifugal pumps must address not only macroscopic performance metrics (flow rate and pressure), but also the microscopic and mesoscopic flow features that govern blood damage. Specifically, minimizing high shear peaks, suppressing recirculation zones, and maintaining sufficient washout even at reduced flow levels were identified as critical design goals.

From this perspective, the potential advantage of a valveless, wave-driven pumping mechanism lies in its ability to generate flow without sustained high rotational speeds and without rigid internal impeller geometries. However, Alex cautioned that such benefits must be demonstrated under realistic circuit conditions and across the full range of clinically relevant flows, including transient and low-flow regimes.

5.4.10 Stage 3: VA versus VV ECMO, Mixing Phenomena, and Directionality of Flow

Distinction between VA and VV configurations (José Sierra). José Sierra then asked Alex to clarify the practical differences between veno–arterial (VA) and veno–venous (VV) ECMO configurations, particularly in terms of flow direction, pressure levels, and the type of complications that typically arise in each case. The question was motivated by the need to understand whether a novel pumping concept could be transferable across configurations, or whether it should be optimized for a specific clinical use case.

Global flow ranges and physiological objectives (Alex). Alex explained that, although the hardware components may appear similar, the physiological objectives of VA and VV ECMO are fundamentally different. In VV ECMO, the primary goal is gas exchange, and typical flow rates are in the range of 2–4 L/min, depending on patient size and lung function. In contrast, VA ECMO provides both circulatory and respiratory support, and therefore often operates at higher flow rates, typically between 3 and 5 L/min, and in some cases even higher.

He emphasized that these values are not fixed, but adjusted continuously according to patient response, venous drainage conditions, and arterial pressure targets.

Pressure levels and suction constraints. With respect to pressures, Alex reiterated that venous-side pressures must remain within a safe window to avoid vessel collapse and hemolysis. Venous inlet pressures around -50 to -60 mmHg are commonly observed, while pressures below approximately -70 mmHg are considered dangerous and trigger alarms or corrective actions.

On the arterial side, outlet pressures can reach several hundred millimeters of mercury, especially in VA ECMO, where the pump must overcome systemic vascular resistance. Alex mentioned pressure differences of 200–300 mmHg across the pump as typical operating values, with transient excursions depending on cannula position and patient condition.

Mixing and differential perfusion in VA ECMO. A critical issue discussed in detail was flow mixing in VA ECMO. Alex explained that, because oxygenated blood is reinjected into the arterial system while the native heart may still be ejecting partially oxygenated blood, complex mixing patterns arise. These patterns depend on:

- the relative strength of the native cardiac output versus the ECMO flow,
- the cannulation site,
- vascular geometry and compliance.

If mixing occurs unfavorably, certain organs (notably the brain and coronary arteries) may receive inadequately oxygenated blood despite apparently sufficient ECMO support. This phenomenon, sometimes referred to clinically as differential hypoxia, represents a major limitation of VA ECMO.

Directionality and flow competition. From a fluid-mechanical perspective, Alex stressed that VA ECMO inherently involves competing flow directions: retrograde flow from the arterial cannula versus antegrade flow generated by the heart. This competition creates stagnation points and shear layers whose position may shift dynamically with changes in heart function.

Such unsteady flow structures are particularly problematic for rotary pumps, as they can induce oscillatory loading, promote recirculation, and increase blood trauma. Alex noted that these effects are difficult to capture using steady-state assumptions, underscoring the importance of transient analysis and time-resolved measurements.

VV ECMO as a more controlled configuration. In contrast, VV ECMO was described as hydraulically simpler, since both drainage and reinfusion occur on the venous side. Although recirculation between drainage and reinfusion cannulas can occur, the absence of arterial reinjection eliminates the strong flow competition seen in VA ECMO.

For this reason, Alex suggested that VV ECMO might be a more suitable initial target for testing alternative pumping concepts, as it allows clearer isolation of pump-induced effects from systemic cardiovascular dynamics.

Implications for pump design and modeling. The discussion concluded with a reflection on how these clinical distinctions translate into engineering requirements. Any pump intended for ECMO must be robust to:

- large pressure gradients,
- transient flow reversals or oscillations,
- operation near suction limits,
- strong coupling with patient-specific vascular dynamics.

Alex emphasized that simplified test loops are necessary for early-stage development, but that eventual validation must address these configuration-dependent phenomena explicitly, especially if VA ECMO is considered as a target application.

5.4.11 Stage 4: Cannulation Strategy, Geometrical Constraints, and Hydraulic Losses

Cannula dimensions and placement (José Sierra). José Sierra next asked for more concrete information regarding the typical cannula sizes used in clinical practice, as well as how their geometry and placement affect achievable flow rates and pressure losses. The motivation was to identify realistic geometrical constraints for any experimental mock-up or numerical model intended to reproduce ECMO operating conditions.

Typical cannula diameters and lengths (Alex). Alex explained that venous drainage cannulas usually have internal diameters in the range of 21–25 Fr (approximately 7–8 mm), while arterial return cannulas are often slightly smaller, typically around 17–19 Fr. These dimensions are selected as a compromise between minimizing insertion trauma and achieving sufficiently low hydraulic resistance.

He emphasized that cannula length is equally important: long cannulas, especially those navigating curved anatomical paths, can introduce significant pressure drops even at moderate flow rates. In practice, the effective hydraulic resistance of the cannulation system may dominate over that of the pump itself.

Pressure losses and flow limitation. Alex highlighted that pressure losses scale nonlinearly with flow rate, and that cannula-induced losses become particularly critical beyond approximately 2 L/min for small-diameter venous cannulas. At higher flow rates, suction pressure at the venous inlet can rapidly drop below safe thresholds, leading to vessel collapse or cavitation-like phenomena in the blood.

He noted that this is often the true limiting factor in ECMO operation: not the pump capacity, but the ability of the venous system to deliver blood without excessive negative pressure.

Cannula positioning and vessel interaction. Beyond purely geometric considerations, Alex stressed that cannula positioning within the vessel plays a crucial role. If the cannula tip is too close to the vessel wall, local flow acceleration can induce partial occlusion, increasing resistance and amplifying suction effects. Small changes in position can therefore produce large variations in measured pressure and flow.

This sensitivity explains why cannulation is considered a critical clinical skill and why pressure monitoring near the cannula tip is indispensable during operation.

Hydraulic asymmetry between inflow and outflow. Another point emphasized by Alex was the inherent asymmetry between drainage and reinfusion lines. While venous drainage is constrained by collapsible vessels and negative pressures, arterial reinfusion occurs against a relatively stiff and pressurized system. As a result, design strategies that work well for the arterial side may be completely unsuitable for the venous side.

This asymmetry has direct implications for experimental test rigs: simply mirroring inlet and outlet geometries does not reproduce physiological conditions.

Implications for experimental mock-ups. In response to a follow-up question, Alex suggested that any laboratory-scale model should include:

- realistic cannula diameters and lengths,
- compliant elements on the venous side to mimic vessel collapse,
- pressure sensors placed close to the cannula tip rather than only at remote locations.

He cautioned that using rigid tubing everywhere can severely underestimate suction-related instabilities and lead to overly optimistic conclusions regarding pump performance.

Transition toward patient-specific considerations. The stage concluded with the observation that cannula selection is patient-specific and depends on vessel size, pathology, and clinical objective. Consequently, Alex argued that any generalized modeling effort should aim to capture trends and failure modes rather than exact patient replication, reserving fine-tuned calibration for later stages of development.

5.4.12 Stage 5: Circuit Volume, Priming, and Transient Operating Regimes

Total circuit volume and clinical relevance (José Sierra). José Sierra then raised the issue of the total blood volume involved in the extracorporeal circuit, asking how much volume is typically displaced from the patient into the system and whether this volume plays a limiting role during operation or during transient phases such as start-up and flow adjustments.

Typical volumes and priming requirements (Alex). Alex explained that the total extracorporeal volume is usually on the order of 0.4–0.6 L, depending on the length and diameter of the tubing, the oxygenator, and any intermediate components such as filters or reservoirs. This volume must be fully primed before connection to the patient, typically using saline or a blood-compatible solution.

He emphasized that, although this volume may appear modest, it is clinically significant, particularly for patients with low circulating blood volume or compromised hemodynamics. Excessive extracorporeal volume can induce dilution effects and transient hypotension at the moment of connection.

Start-up and ramping protocols. Alex stressed that flow is never applied abruptly. Instead, ECMO systems are brought online using controlled ramping procedures, gradually increasing the flow rate while continuously monitoring venous pressure, arterial pressure, and patient response. Sudden increases in flow can cause rapid drops in venous pressure, leading to suction events or vessel collapse.

In practice, clinicians often increase flow in steps of a few hundred milliliters per minute, allowing the system to equilibrate before proceeding further.

Transient pressure fluctuations. During these transient phases, pressure signals may exhibit significant oscillations, even when the mean flow remains moderate. Alex noted that these oscillations are not necessarily pathological but reflect the interaction between pump dynamics, tubing compliance, and vascular response.

However, if negative pressure excursions exceed safe thresholds (typically around -70 mmHg), the ramping process must be halted or reversed.

Role of compliance and damping. Alex highlighted that compliant elements in the circuit—both intentional (e.g., compliant tubing) and physiological (venous vessels)—act as hydraulic capacitors, smoothing pressure fluctuations during transients. Rigid systems, by contrast, tend to amplify pressure spikes and are therefore less forgiving during flow changes.

This observation reinforces the importance of incorporating compliance into experimental mock-ups if transient behavior is to be meaningfully reproduced.

Implications for modeling and experimentation. From a modeling perspective, Alex suggested that steady-state analyses alone are insufficient. Even if a configuration appears acceptable under steady conditions, transient simulations may reveal unsafe pressure excursions or instabilities.

For this reason, he recommended that any numerical or experimental study should explicitly consider:

- start-up and shut-down sequences,
- flow ramping rates,
- transient pressure extrema rather than only mean values.

Closing remarks on system-level behavior. Alex concluded this part of the discussion by emphasizing that ECMO performance must be assessed at the system level. Pump characteristics, cannula geometry, tubing compliance, and patient response are tightly coupled, and isolating one component without considering the others can lead to misleading conclusions.

5.4.13 Stage 6: Safety Margins, Alarm Logic, and Clinical Decision-Making

Alarm thresholds and their clinical interpretation (José Sierra). José Sierra asked how clinicians interpret pressure and flow alarms in practice, and whether these thresholds are rigid limits or context-dependent indicators. The question aimed to clarify how much operational margin exists between nominal operating conditions and situations considered clinically unsafe.

Venous-side suction alarms and corrective actions (Alex). Alex explained that venous-side negative pressure alarms are among the most critical in ECMO operation. As discussed previously, a commonly used alarm threshold is around -70 mmHg at the drainage side. When this threshold is reached, the system may respond in different ways depending on the pump console and institutional protocol:

- automatic reduction of pump speed to relieve suction,

- visual and acoustic alarms prompting immediate clinical intervention,
- temporary stabilization while venous return is reassessed.

Importantly, Alex stressed that these thresholds are not arbitrary: sustained operation beyond them risks vessel collapse, air entrainment, and severe hemolysis. Nevertheless, short excursions close to the limit can occur transiently during patient movement or flow adjustments and are not necessarily catastrophic if promptly corrected.

Arterial-side pressure alarms. On the arterial return side, positive pressure alarms are typically set around 400 mmHg. Exceeding this level suggests excessive downstream resistance, cannula obstruction, or kinking of the line. In such cases, increasing pump speed does not increase effective perfusion and instead raises the risk of circuit damage or patient harm.

Alex emphasized that arterial alarms often indicate a mechanical or geometric issue rather than a need for more pumping power.

Flow-based alarms and hidden failure modes. A critical point raised by Alex was that flow-based alarms alone can be misleading. Modern centrifugal pumps may maintain an apparently stable flow reading while progressively reducing rotational speed in response to suction signals. If the clinician focuses only on displayed flow values without considering pressure trends, a gradual loss of effective support may go unnoticed.

For this reason, Alex advocated for interpreting flow and pressure signals together, rather than relying on any single indicator.

Decision rules under unstable conditions. When confronted with unstable pressure–flow behavior, clinicians typically follow a hierarchy of corrective actions:

1. reduce pump speed to alleviate suction or overpressure,
2. reassess cannula position and patient volume status,
3. administer fluids or reposition the patient if venous return is insufficient,
4. in persistent cases, change cannula size or configuration.

Alex noted that increasing pump power is rarely the correct response to instability, especially on the venous side.

Implications for alternative pump concepts. In the context of evaluating non-rotary, valveless pumping concepts, Alex emphasized that safety logic must be considered from the outset. Any alternative device must:

- provide reliable indicators of suction and overpressure,
- fail safely under adverse conditions,
- allow clinicians to recognize and correct instability without ambiguity.

He cautioned that even if a new pump exhibits favorable hemocompatibility characteristics, it will not be clinically viable unless its behavior under off-nominal conditions is transparent and predictable.

Closing of the clinical discussion. The discussion concluded with a shared understanding that clinical acceptance depends not only on mean performance metrics, but on how the system behaves near its limits. Safety margins, alarm strategies, and interpretability of signals were identified as essential criteria for guiding both experimental validation and future device design.

6 Conclusions

This document has compiled and structured the technical outcomes of the meetings held in Valladolid in February 2026, encompassing laboratory observations, post-laboratory strategic discussions, and a clinically oriented exchange at Hospital Clínico Universitario de Valladolid. Together, these activities provided a coherent and multi-perspective foundation for the development of an alternative, valveless pumping concept for extracorporeal circulation.

Laboratory and Experimental Perspective

The laboratory visit enabled direct inspection of the experimental infrastructure developed at the University of Valladolid, including the fabrication of highly compliant tubes, the actuation mechanisms used to generate wave-driven flow, and the associated instrumentation for pressure, flow, and high-speed imaging. These observations clarified the practical feasibility of manufacturing compliant conduits with tailored mechanical properties and highlighted the central role of large-deformation behavior, contact, and fluid–structure interaction in Liebau-based pumping mechanisms.

The experimental setup was identified as sufficiently flexible to support a staged experimental program, progressing from structural-only tests to fully coupled fluid–structure interaction and, ultimately, to system-level configurations incorporating rigid and compliant segments.

Post-Laboratory Strategic Alignment

The post-laboratory meeting served to consolidate the experimental insights into a structured roadmap for future work. A phased strategy was agreed upon, emphasizing progressive complexity to ensure numerical robustness and physical interpretability. Key methodological conclusions include the necessity of defining an initial prestressed equilibrium configuration, careful treatment of contact geometry, and the separation of local high-fidelity modeling from global system representations.

The role of reduced-order models (ROMs) was clearly positioned as an acceleration and exploration tool built upon validated full-order simulations, rather than as a direct replacement for high-fidelity analysis in the early stages.

Clinical Context and Reference Technology

The visit to Hospital Clínico provided essential clinical grounding for the project. The discussion with the clinician clarified realistic operating envelopes for extracorporeal circulation, including typical adult flow rates (2–7 L/min), pressure rises (200–300 mmHg, with alarm thresholds up to $\sim$ 400 mmHg), and critical venous suction limits (alarm around -70 mmHg). Cannula dimensions, circuit volumes, priming requirements, and transient operating procedures were also detailed.

Current centrifugal ECMO pumps were explicitly identified as the technological baseline. Their advantages in reliability and compactness were contrasted with well-known limitations, such as sustained high shear exposure, recirculation zones, hemolysis, and flow-dependent thrombogenicity, particularly in low-flow regimes. These limitations define the clinical motivation for exploring alternative pumping concepts.

Implications for Design and Modeling

Across all discussions, a consistent conclusion emerged: performance metrics such as mean flow and pressure are necessary but insufficient criteria for evaluating extracorporeal pumps. Hemocompatibility, robustness under transient conditions, sensitivity to venous preload, and clarity of failure modes are equally critical.

From an engineering standpoint, the meetings translated clinical practice into explicit design constraints and modeling requirements. Any alternative pump must:

- operate reliably within the clinically relevant flow and pressure ranges,
- avoid excessive negative pressures on the venous side,
- minimize high shear peaks and recirculation zones,
- behave predictably and safely near operational limits.

The valveless, wave-driven pumping concept investigated in this project was recognized as a promising candidate in this respect, provided that its performance and safety can be demonstrated under realistic circuit conditions.

Overall Outcome

In summary, the Valladolid meetings established a shared understanding between experimentalists, modelers, and clinicians. They transformed exploratory ideas into a technically grounded research plan, anchored in real clinical practice and informed by both experimental feasibility and numerical tractability. The conclusions drawn here provide a solid basis for advancing toward validated simulations, refined experimental prototypes, and, ultimately, a credible alternative to current rotary ECMO pump technology.

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